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CADILLAC AREA GROUNDWATER CONTAMINATION
FEASIBILITY STUDY
CADILLAC, MICHIGAN

INTERIM DELIVERABLE NO. 1

EPA Region 5 Records Ctr.



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INTERIM DELIVERABLE NO. 1

Prepared for:
MICHIGAN DEPARTMENT OF NATURAL RESOURCES

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CADILLAC AREA GROUNDWATER CONTAMINATION
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TABLE OF CONTENTS

Section	Title	Page No.
1.0	INTRODUCTION.	1-1
1.1	PURPOSE AND SCOPE.	1-2
1.2	SITE DESCRIPTION	1-3
2.0	IDENTIFICATION OF REMEDIAL ACTION OBJECTIVES.	2-1
2.1	GENERAL RESPONSE OBJECTIVES.	2-2
2.2	SUMMARY OF CHEMICAL-SPECIFIC APPLICABLE OR RELEVANT AND APPROPRIATE REQUIREMENTS	2-4
2.3	TARGET LEVELS FOR GROUNDWATER REMEDIATION.	2-8
2.3.1	Approach.	2-8
2.3.2	Development of Preliminary Groundwater Target Levels.	2-11
2.3.2.1	Shallow Volatile Organic Compound Plume.	2-12
2.3.2.2	Intermediate Volatile Organic Compound Plume.	2-16
2.3.2.3	Rexair Plume	2-16
2.3.2.4	Northernair Chromium Plume.	2-21
2.3.3	Final Groundwater Target Levels	2-21
2.4	TARGET LEVELS FOR SOIL REMEDIATION	2-21
2.4.1	Approach.	2-23
2.4.2	Development of Preliminary Soil Target Levels	2-24
2.4.2.1	Kysor.	2-24
2.4.2.2	Four Winns	2-24
2.4.3	Final Soil Target Levels.	2-24
2.5	TREATED GROUNDWATER DISCHARGER REQUIREMENTS.	2-27

GLOSSARY OF ACRONYMS

APPENDICES

APPENDIX A - CHEMICAL-SPECIFIC ARARS

APPENDIX B - CALCULATIONS FOR PRELIMINARY TARGET LEVELS

LIST OF TABLES

Table	Title	Page No.
2-1	CADILLAC AREA GROUNDWATER STUDY: GROUNDWATER CHEMICAL-SPECIFIC ARARS COMPARISON.	2-7
2-2	CADILLAC AREA GROUNDWATER CONTAMINATION: PRELIMINARY TARGET LEVELS, SHALLOW PLUME.	2-13
2-3	RESIDUAL RISK ASSOCIATED WITH TARGET CLEAN-UP LEVELS. . .	2-14
2-4	CADILLAC AREA GROUNDWATER CONTAMINATION: PRELIMINARY TARGET LEVELS, INTERMEDIATE PLUME	2-17
2-5	RESIDUAL RISK ASSOCIATED WITH TARGET CLEAN-UP LEVELS. . .	2-18
2-6	CADILLAC AREA GROUNDWATER CONTAMINATION: PRELIMINARY TARGET LEVELS, REXAIR PLUME	2-20
2-7	RESIDUAL RISK ASSOCIATED WITH TARGET CLEAN-UP LEVELS. . .	2-22
2-8	CADILLAC AREA GROUNDWATER CONTAMINATION: PRELIMINARY TARGET LEVELS, KYSOR SOILS.	2-25
2-9	CADILLAC AREA GROUNDWATER CONTAMINATION: PRELIMINARY TARGET LEVELS, FOUR WINNS SOILS	2-26
2-10	TREATED GROUNDWATER DISCHARGE REQUIREMENTS.	2-28

LIST OF FIGURES

Figure	Title	Page No.
2-1	SHALLOW PLUME CONTAMINATION	2-15
2-2	INTERMEDIATE PLUME CONTAMINATION.	2-19

1.0 INTRODUCTION

Under contract to the Michigan Department of Natural Resources (MDNR), E.C. Jordan Co. (Jordan) is presently conducting a Remedial Investigation/Feasibility Study (RI/FS) of the groundwater contamination below the industrial park in Cadillac, Michigan. Because the Northernnaire Plating Company (Northernnaire) and Kysor of Cadillac, Inc. (Kysor) Superfund sites are located within the study area, the study is being conducted under joint MDNR-U.S. Environmental Protection Agency (USEPA) guidance, and in accordance with the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA), as amended by the Superfund Amendments and Reauthorization Act of 1986 (SARA), and the National Oil and Hazardous Substances Pollution Contingency Plan (NCP 1985 - Final Rule, Section 300.68). The Northernnaire site and associated metals contamination were the subjects of an RI/FS, also performed by Jordan for MDNR. Because the Northernnaire plume was known to mingle with other source contaminants, it was decided that the Northernnaire operable unit for the groundwater would be performed in conjunction with the Cadillac FS, which also includes assessments of volatile organic compound (VOC) contamination from Kysor and other identified sources in the Cadillac Industrial Park.

Investigation and remediation of site contamination for the Cadillac site has been divided into two parts. The RI presented the distribution of VOC and chromium contamination in groundwater beneath the industrial park of Cadillac and adjacent areas of Haring Township, Michigan, and identified residual contaminants in soils at suspected source locations. Information sources for

the RI were work performed by Jordan, MDNR, and USEPA, and by consultants for industries within the industrial park. A draft baseline risk assessment (RA), now under review, was then prepared based on the findings of the RI. The FS will identify and analyze alternatives for groundwater remediation of the contaminated plumes and soil remediation of the identified source locations.

1.1 PURPOSE AND SCOPE

Based on the results of the RI and RA, this interim deliverable presents the remedial action objectives for the Cadillac FS, including preliminary target levels, and treated groundwater discharge requirements. Upon MDNR-USEPA concurrence of the remedial action objectives for the Cadillac site, Jordan will conduct the following tasks to complete the FS:

- o identification of final target levels;
- o identification and screening of remedial technologies;
- o development of alternatives;
- o initial screening of alternatives; and
- o detailed analysis of alternatives.

This FS will serve as the basis for the Record of Decision (ROD), which documents remedy selection, and for remedial design and construction.

1.2 SITE DESCRIPTION

The industrial park area in Cadillac, Michigan, includes nearly 40 industrial facilities. Past uncontrolled releases of contaminants to the ground by several of these facilities have contaminated groundwater in aquifers underlying the park. The contaminants of concern are VOCs (i.e., chlorinated and aromatic hydrocarbons such as trichloroethylene [TCE] and xylene) and hexavalent chromium (derived from chromium plating wastes). The presence of contaminants within the upper aquifers beneath the site has created concern for potential impact on a lower aquifer, which supplies potable water to the city from a wellfield located near the center of the industrial park.

The RI identified five contaminant sources and seven contaminant plumes in groundwater in the study area. The five probable or known source locations are (1) Northernnaire; (2) Kysor; (3) Rexair (probable); (4) Four Winns; and (5) Four Star/Joynts (former).

The seven plumes are categorized under four major plumes and include the following:

Northernnaire Plumes

1. A hexavalent-chromium plume in the intermediate aquifer situated just above the lower clay aquitard. Northernnaire was the source.
2. A small hexavalent-chromium plume of relatively low concentration in the shallow aquifer, interpreted as a small slug released from Northernnaire, which, moving rapidly through the upper aquifer and being diluted as it migrates, should soon merge with the intermediate aquifer.

Shallow Aquifer Plume

A large, concentrated VOC plume (up to 115 parts per million [ppm]) in the shallow aquifer caused primarily by past solvent dumping behind Kysor. This plume has merged with other VOC releases from Four Star/Joynts (former) to the northeast and Four Winns to the northwest.

Intermediate Aquifer Plume

A large, but less concentrated, VOC plume (up to 12 ppm) in the intermediate aquifer resulting from Kysor-related VOCs migrating from the shallow aquifer through the upper aquitard.

Rexair Plume

A highly concentrated VOC plume (up to 153 ppm) at a depth of 140 feet, located in the intermediate aquifer between Rexair and the city wellfield.

Other Plumes

The RI also identified two small VOC plumes in the North Park Subdivision. These plumes are located horizontally downgradient of the city wellfield and appear to pose a less immediate threat (if any) to the city water supply. It is believed that all potential receptors (i.e., domestic wells) downgradient of these plumes were replaced by municipal water supplies. Therefore, these plumes were not assessed as a potential human health risk in the baseline RA.

Only one area of soils, the former solvent disposal area behind Kysor, was positively identified in the RI as having significant residual VOC contamination. This area appears to be a continuing source of groundwater contamination in the shallow and intermediate aquifers west of the city wellfield. In addition, soils at Four Winns are assumed to be a potential continuing source of groundwater contamination. However, because Four Winns, Inc. elected to participate as a potentially responsible party in the study with its own consultant, Jordan was not able to assess soil contamination at Four Winns as a source of groundwater contamination; but, based on 1985 data from MDNR, Jordan has assumed that concentration levels in Four Winns soils could require remediation. Other sources of soils contamination could be at the Ingraham property and at Rexair. Studies further defining both plumes and sources at

Rexair are being conducted by a consultant engaged by Rexair. Other explorations at Ingraham may be performed by MDNR under another contract to MDNR.

2.0 IDENTIFICATION OF REMEDIAL ACTION OBJECTIVES

Remedial action objectives and site-specific considerations must be identified to select the appropriate remedial technologies and formulate remedial alternatives. Remedial action objectives are statements of the proposed level of cleanup associated with the specific chemical contamination problems in the various media at a site. Remedial action objectives are aimed at protecting human health and the environment and are based on the contaminant(s) of concern, exposure route(s), and receptor(s); they are acceptable contaminant levels or ranges of levels for each exposure route. Applicable or Relevant and Appropriate Requirements (ARARs) that establish clean-up standards are also used to determine remedial action objectives.

Section 2.1 identifies the general response objectives based on the results of the RA and other considerations in the NCP 40 CFR 300.68(e)(2). Chemical-specific ARARs, previously identified in the RI, are summarized in Section 2.2. These ARARs are used in conjunction with the general objectives to formulate specific target levels for groundwater and soil remediation. These target levels are presented in Sections 2.3 and 2.4. Discharge requirements for potential discharges of treated groundwater to surface water, publicly owned treatment works (POTWs), or recharge to groundwater are presented in Section 2.5; the requirements are used to determine the appropriate level of treatment before discharge.

2.1 GENERAL RESPONSE OBJECTIVES

In the RA, the only exposure pathway identified as presenting a potential threat to public health is the domestic use of contaminated groundwater, including drinking, showering, bathing, and cooking. Because each contaminated groundwater plume identified and evaluated in the RA is considered an existing or potential source of drinking water, the following separate remedial action objectives are proposed for each plume (i.e., the shallow plume, intermediate plume, Northernnaire plume, and the Rexair plume):

Shallow VOC Plume. Reduce human health risks posed by potential use of the shallow groundwater plume as a domestic water supply (including drinking). The shallow plume is contaminated with 1,1,1-trichloroethane (1,1,1-TCA); 1,1-dichloroethylene (1,1-DCE); trichloroethylene (TCE); 1,2-dichloroethane (1,2-DCA); and tetrachloroethylene (PCE).

Intermediate VOC Plume. Reduce human health risks posed by the potential use of the intermediate groundwater plume (which is contaminated with trans-1,2-dichloroethylene [t-1,2-DCE]; TCE; 1,1-DCE, and PCE) as a domestic drinking water supply.

Northernnaire Plumes. Reduce human health risks posed by the potential use of the hexavalent chromium-contaminated groundwater in the Northernnaire Plumes as a domestic drinking water supply.

Rexair Plume. Reduce potential human health risks posed by using the Rexair groundwater plume, primarily contaminated with TCE and methylene chloride (MC), as a domestic drinking water supply.

As discussed in the RA, the residual soil contamination at Kysor and Four Winns does not pose a direct contact threat because the contaminated soil is subsurface. However, the RA did assess residual soil contamination at Kysor as a continuing source of groundwater contamination, based on the concentrations of the contaminants and their ability to leach to groundwater. Adequate data were not available to assess existing soil contamination at Four Winns as a source of groundwater contamination. However, based on 1985 data, it is assumed that concentration levels at Four Winns are high enough to warrant remediation. Based on this assessment, the following remedial action objectives are proposed:

Kysor. Reduce potential human health risks posed by the continued leaching of TCE, toluene, and 1,1,1-TCA from subsurface soils at Kysor to a potential drinking water source. Reduce potential human health risks posed by the future potential for xylene to leach from subsurface soils at Kysor to a potential drinking water source.

Four Winns. Reduce potential human health risks posed by the potential for continued leaching of 1,1,1-TCA, toluene, TCE, and MC from subsurface soils at Four Winns to a potential drinking water source.

2.2 SUMMARY OF CHEMICAL-SPECIFIC APPLICABLE OR RELEVANT AND APPROPRIATE REQUIREMENTS

As discussed in the Cadillac RI, ARARs are federal and state public health and environmental requirements and guidelines that apply to hazardous waste site cleanup. Section 121 of CERCLA, amended by SARA, requires that hazardous waste remedial actions comply with federal ARARs and state ARARs that are more stringent, legally enforceable, and consistently enforced statewide. ARARs are used to determine the appropriate extent of site cleanup, to scope and formulate remedial action alternatives, and to govern the implementation and operation of the selected action.

Applicable requirements and relevant and appropriate requirements are defined as follows:

- o Applicable Requirements are those clean-up standards, standards of control, or other substantive environmental requirements, criteria, or limitations promulgated under federal or state law that specifically address a hazardous substance at a CERCLA site. An example of an applicable requirement is the use of Maximum Contaminant Levels (MCLs), promulgated under the Safe Drinking Water Act (SDWA), for a site where groundwater contamination is entering a public water supply.
- o Relevant and Appropriate Requirements are federal- and state-promulgated clean-up standards, standards of control, or other substantive environmental protection requirements, criteria, or limitations that, while not

legally applicable, can be applied if, in the decision-maker's best professional judgement, site circumstances are similar to jurisdictionally covered situations, and use of the requirements makes good sense. The term "relevant" is included so that requirements initially screened out as non-applicable because of jurisdictional restrictions, can be reconsidered. For example, MCLs would be relevant and appropriate requirements for use at a site where groundwater contamination could effect a potential, as opposed to an actual, drinking water source. Relevant and appropriate requirements should be given the same weight for consideration as applicable requirements.

- o To be Considered Materials (TBCs) are federal and state non-promulgated advisories or guidance documents that do not have ARAR status. However, if there are no specific ARARs for a chemical or situation, or if existing ARARs are not deemed sufficiently protective, then guidance or advisory criteria should be identified and used to better ensure public health and environmental protection.

As discussed in the RI, ARARs may be characterized as chemical-, location-, or action-specific. Chemical-specific requirements are used to determine the remedial action objectives because they set health- or risk-based concentration limits or discharge limitations in various environmental media for specific hazardous substances, pollutants, and contaminants. They govern the extent of site remediation by providing either actual clean-up levels or a basis for calculating such levels. If a chemical has more than one requirement that is an ARAR, the most stringent generally should be attained. If no ARAR exists,

or if the ARAR for a substance is insufficient to be protective, then the federal or state TBC should be used in conjunction with the RA to determine the appropriate clean-up level.

Chemical-specific ARARs for the Cadillac site were identified and discussed in the RI. ARARs that set concentration limits for groundwater contaminants include SDWA MCLs, SDWA Maximum Contaminant Level Goals (MCLGs), and Federal Ambient Water Quality Criteria (AWQC). In addition, Michigan is presently drafting proposed Groundwater Quality Rules. TBCs include USEPA Health Advisories, USEPA Risk Reference Doses, and USEPA Carcinogen Assessment Group Potency Factors. No federal or state regulatory requirements currently govern the extent of cleanup for contaminants in soil. Chemical-specific ARARs that set limitations for discharges to surface water and POTWs were also discussed in the RI and are addressed in Section 2.5. A list of all chemical-specific ARARs and TBCs for the Cadillac site and a synopsis of the requirements are contained in Appendix A. Actual required concentration levels for the groundwater, which are used to set groundwater remedial action objectives, are listed in Table 2-1, and are compared to the maximum concentrations of the groundwater chemicals identified as contributing the most to total risk in each plume. Because contaminants in the soil are presently leaching or have the future potential to leach to groundwater, concentrations of chemicals which are presently not a problem in groundwater, but which are found at high concentrations in soil, are also compared to groundwater ARARs.

TABLE 2-1
CADILLAC AREA GROUNDWATER STUDY: GROUNDWATER
CHEMICAL-SPECIFIC ARARs COMPARISON

Compound	ARARs (mg/ℓ)			Existing Maximum Concentrations (mg/ℓ)			
	MCL	MCLG	AWQC*	Rexair	Upper Aquifer	Intermediate Aquifer	Northernaire Plume ¹
1,1,1-trichloroethane	0.2	0.2	18.4	.017	32	1.2	0.44
trans-1,2-dichloroethylene	--	0.07**	--	.048	--	4.6	.04
1,1-dichloroethylene	0.007	0.007	0.00033	--	1.0	.099	.026
1,2-dichloroethane	0.005	0	0.00094	--	2.0	--	--
methylene chloride	--	--	0.00019	.078	0.69	0.24	--
tetrachloroethylene	--	0**	0.0008	--	0.72	0.17	0.003
trichloroethylene	0.005	0	0.0028	153	78	7.5	2.5
chromium (hexavalent)	0.05	0.12**	0.05	--	--	--	0.773
xylene	--	0.44**	--	--	1.7	--	--
toluene	--	2.0**	14.3	.042	3.2	.001	--

* AWQC are adjusted for drinking water exposure only (USEPA, 1986) or calculated for water and fish ingestion if drinking water exposure value does not exist.

** Proposed levels.

¹ The VOCs present within the boundaries of the Northernaire plume are not derived from Northernaire; the chromium from Northernaire and VOCs from other sources have mixed in the aquifer.

2.3 TARGET LEVELS FOR GROUNDWATER REMEDIATION

Groundwater target levels are the chemical concentrations in each plume that would be protective of human health if long-term exposures were to occur. The baseline RA for the Cadillac site (Chapter 8 of the RI) provides a basis for determining which chemicals of concern should be targeted for cleanup in each plume. It is not necessary to calculate target levels for every chemical found in the plumes. Cleanup should focus on those chemicals that have been shown in the RA to pose "significant" risks to public health (i.e., carcinogenic risks within or above the target range of 10^{-4} to 10^{-7} , or noncarcinogenic risks greater than one). This section describes the approach to development of groundwater target levels and presents the target levels for each plume.

2.3.1 Approach

The first step in the development of target levels for groundwater remediation at Cadillac is to identify the chemicals in each plume that present "significant" risks in the baseline RA (as defined previously, carcinogenic risks within or above the target range or noncarcinogenic risk greater than one). For the Cadillac area groundwater contamination, these are as follows:

- o Shallow VOC Plume:

- 1,1,1-TCA

- 1,1-DCE

TCE

1,2-DCA

PCE

o Intermediate VOC Plume:

1-1,2-DCE

TCE

1,1-DCE

PCE

o Rexaire Plume:

TCE

MC

o Northernnaire Chromium Plume

Hexavalent Chromium (Cr[+6])

Preliminary target levels are calculated for these chemicals on the basis of health-based ARARs, or if ARARs do not exist, public health risk calculations.

All risk calculations can be found in Appendix B. For carcinogens, target levels are developed to fall within a total risk range of 10^{-4} to 10^{-7} because, as stated in the Superfund Public Health Evaluation Manual (USEPA, 1986), site remedies should reduce ambient chemical concentrations to levels within that range when possible. The formulas used to calculate preliminary target levels for those carcinogens that have no health-based ARARs are as follows:

$$\begin{array}{lcl} \text{Target Risk for} & & \text{Target Risk Level (10}^{-4} \text{ to 10}^{-7}\text{)} \\ \text{Individual} & = & \text{-----} \\ \text{Compound} & & \text{Total Number of Chemicals} \end{array}$$

(The 10⁻⁶ target risk level will be used in the preliminary target level calculations for illustration.)

$$\begin{array}{lcl} \text{Target Chronic Daily Intake} & = & \text{Target Risk} \\ \text{(CDI, mg/kg/day)} & & \text{CAG Potency Factor (mg/kg/day)}^{-1} \end{array}$$

$$\begin{array}{lcl} \text{Target Level (mg/l)} & = & \text{Target CDI (mg/kg/day)} \\ & & \text{Human Intake Factor (l/kg/day)} \end{array}$$

WHERE:

$$\begin{array}{lcl} \text{Human Intake Factor} & = & \text{Amount of Water Consumed per Day} \\ & & \text{Average Adult Body Weight} \\ & = & \frac{2 \text{ l/day}}{70 \text{ kg}} = 0.029 \end{array}$$

Once preliminary target levels for carcinogens are identified (either from ARARs or risk calculations), the residual risks that would be associated with lifetime ingestion of the groundwater that contains chemicals at the target levels are calculated. In some cases, the residual risks exceed the target range. In these cases, health-based target levels are calculated for some of the chemicals whose ARARs exceed the target range, and residual risks are recalculated.

For noncarcinogens that do not have ARARs, target levels are calculated so that the resultant risk ratio is less than one. The following equations are used:

$$\text{Target Risk for Individual Compound} = \frac{1}{\text{Total Number of Chemicals}}$$

$$\text{Target Chronic Daily Intake (CDI, mg/kg/day)} = \text{Target Risk} \times \text{RfD (mg/kg/day)}$$

$$\text{Target Level (mg/l)} = \frac{\text{Target CDI (mg/kg/day)}}{\text{Human Intake Factor (l/kg/day)}}$$

(Human intake factor is the same as for carcinogens.)

Preliminary target clean-up levels (TCLs) provide options that are used in the FS to assist in the selection and screening of remedial technologies and in the analysis of risks for remedial alternatives. The preliminary target levels are revised in the Final Target Level subsection in response to engineering, cost, policy, and/or environmental considerations. Final target levels will, therefore, achieve a specified total target risk but may also reflect public health, environmental, engineering, policy, and cost considerations. This approach, which integrates several considerations, allows for the most efficient reduction of risk.

2.3.2 Development of Preliminary Groundwater Target Levels

The following subsections summarize the identification of preliminary target levels for groundwater contaminants in each plume.

2.3.2.1 Shallow Volatile Organic Compound Plume. Six chemicals of concern in the shallow VOC plume were identified in the RA as presenting potentially significant risks to public health. These include two noncarcinogens and four carcinogens. Of these chemicals, all have health-based ARARs available. However, PCE only has an AWQC (0.0008 mg/l) which is below detection limits. Therefore, a TCL corresponding to the 10^{-6} risk level was calculated for PCE (see Appendix B). The preliminary TCLs for the shallow plume contaminants can be found in Table 2-2.

The residual risks associated with the preliminary target levels for the four carcinogens in the shallow plume are calculated and presented in Table 2-3 (Option 1). The total risk associated with the plume at preliminary TCLs is 1.34×10^{-4} , most of which is associated with exposure to 1,1-DCE. This is quite high for a potential drinking water source. Therefore, the target levels that would be associated with a total risk of 1×10^{-6} were calculated and are also presented in Table 2-3 (Option 2). These target levels are extremely low (ranging from 0.015 ug/l for 1,1-DCE to 0.8 ug/l for TCE), and are below the detection limits for these chemicals. Therefore, a third option was considered, where the preliminary TCLs for TCE, 1,2-DCA, and PCE remain as in Option 1 but the target level for 1,1-DCE is adjusted to the 10^{-6} risk level. The total residual risk associated with Option 3 is 1.77×10^{-5} . This is within the target risk range. The residual risks associated with each of these options are also presented graphically in Figure 2-1. Final target levels for the shallow plume are identified in Section 2.3.3.1.

TABLE 2-2
 CADILLAC AREA GROUNDWATER CONTAMINATION
 PRELIMINARY TARGET LEVELS
 SHALLOW PLUME

<u>CHEMICAL</u>	<u>(mg/l)</u>	<u>SOURCE</u>
1,1,1-TCA	0.2	MCL
TOLUENE	2	MCLG
1,1-DCE	0.007	MCL
TCE	0.005	MCL
1,2-DCA	0.005	MCL
PCE	0.001	CALCULATED

TABLE 2-3

Residual Risk Associated with Target Cleanup Levels
Cadillac Area Groundwater Contamination
Ground Water Ingestion

SHALLOW PLUME: OPTION 1

Residual Risks Associated with Groundwater Contaminants at their Preliminary Target Levels

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAG Potency Estimate (ug/kg/day) ⁻¹	Risk Estimate
1,1-DCE	7.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.00E-01	1.00E+03	5.90E-01	1.16E-04
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
1,2-DCA	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.55E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	9.10E-02	1.30E-05
PCE	1.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-06
											TOTAL RISK	1.32E-04

SHALLOW PLUME: OPTION 2

Calculation of Target Levels at 10⁻⁶ Risk Level

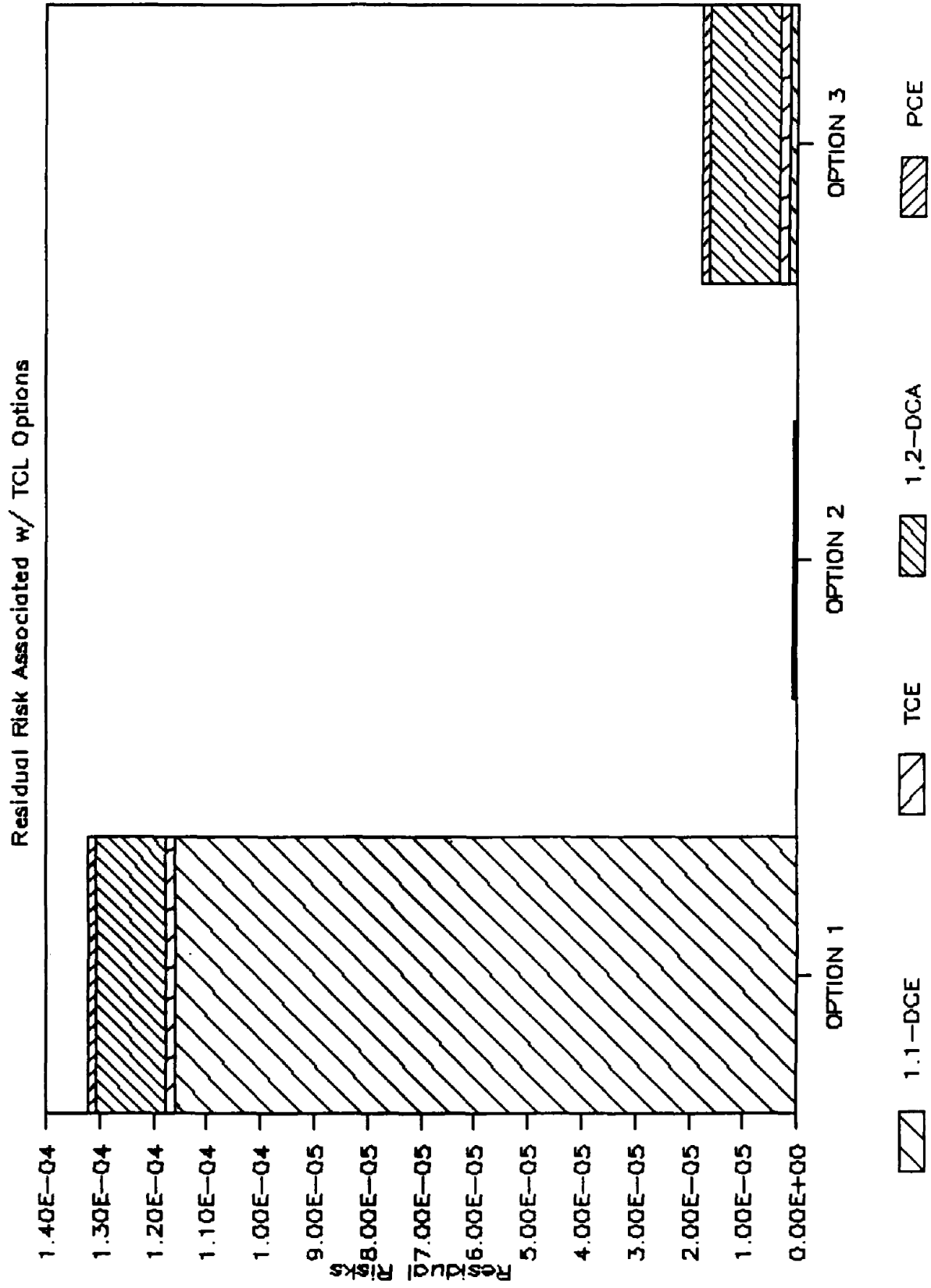
Compound	AF	Target Risk	Potency Factor (ug/kg-day) ⁻¹	Target CDI (ug/kg/day)	Amount Contacted (l/day)	Amount Ingested (l/day)	Dermal Absorption Fraction	Ingestion Absorption Fraction	Frequency of Contact (days)	Fraction of Year Exposed	Years of Exposure	Fraction of Lifetime Exposed	Body Weight (kg)	Dermal Intake (l/kg/day)	Ingestion Intake (l/kg/day)	Target Level
1,1-DCE	2.50E-01	2.50E-07	5.80E-01	4.31E-07	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.86E-02	11.51E-02
TCE	2.50E-01	2.50E-07	1.13E-02	2.27E-05	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.86E-02	17.95E-01
1,2-DCA	2.50E-01	2.50E-07	9.10E-02	2.75E-06	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.86E-02	19.62E-02
PCE	2.50E-01	2.50E-07	5.20E-02	4.81E-06	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.86E-02	11.65E-01

SHALLOW PLUME: OPTION 3

Residual Risks Associated with: TCE and 1,2-DCA at MCLs; 1,1-DCE and PCE at 10⁻⁶ Risk Level

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAG Potency Estimate (ug/kg/day) ⁻¹	Risk Estimate
1,1-DCE	1.00E+01	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.86E-02	1.00E+03	5.90E-01	1.55E-06
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
1,2-DCA	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.55E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	9.10E-02	1.30E-05
PCE	1.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-06
											TOTAL RISK	1.72E-05

FIGURE 2-1. SHALLOW PLUME CONTAMINATION



2.3.2.2 Intermediate Volatile Organic Compound Plume. Four chemicals of concern in the intermediate VOC plume were identified in the RA as presenting potentially significant risks to public health. These include one noncarcinogen and three carcinogens. Of these chemicals, as in the shallow plume, only PCE does not have an applicable health-based ARAR which is above practical detection limits. Therefore, the PCE concentration which corresponds to the 10^{-6} risk level was calculated in Appendix B. Preliminary target levels for contaminants in the intermediate plume are presented in Table 2-4.

Residual risks associated with preliminary target levels were also calculated for the intermediate plume in a similar manner to the shallow plume. The risks associated with Options 1, 2, and 3 are presented in Table 2-5 and depicted graphically in Figure 2-2. As with the shallow plume, most of the residual risk is posed by 1,1-DCE, so its target level was adjusted in Option 3. Final target levels for the intermediate VOC plume are identified in Section 2.3.3.2.

2.3.2.3 Rexair Plume. Although most of the risk posed by the Rexair plume is associated with TCE, MC also presented a risk within or above the target range for several exposure scenarios in the RA. Therefore, preliminary target levels were identified for both chemicals.

TCE has an applicable ARAR, a final MCL of 0.005 mg/l. However, MC only has an AWQC (0.00019) which is far below practical detection limits. Therefore, the MC concentration that corresponds to the 10^{-6} risk level is calculated in Appendix B. Preliminary target levels for the Rexair plume are presented in Table 2-6.

TABLE 2-4
 CADILLAC AREA GROUNDWATER CONTAMINATION
 PRELIMINARY TARGET LEVELS
 INTERMEDIATE PLUME

<u>CHEMICAL</u>	<u>(mg/l)</u>	<u>SOURCE</u>
T-1, 2-DCE	0.07	MCLG
TCE	0.005	MCL
1,1-DCE	0.007	MCL
PCE	0.001	CALCULATED

TABLE 2-5

Residual Risk Associated with Target Cleanup Levels
Cadillac Area Groundwater Contamination
Ground Water Ingestion

INTERMEDIATE PLUME: OPTION 1

Residual Risks Associated with Groundwater Contaminants at their Preliminary Target Levels

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAB Potency Estimate (ug/kg/day)-1	Risk Estimate
1,1-DCE	7.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.00E-01	1.00E+03	5.80E-01	1.15E-04
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
PCE	1.00E-03	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-05	1.00E+03	5.20E-02	1.49E-09
											TOTAL RISK	1.18E-04

INTERMEDIATE PLUME: OPTION 2

Calculation of Target Levels at 10⁻⁶ Risk Level

Compound	AF	Target Risk	Potency Factor (mg/kg/day)	Target CDI (ug/kg/day)	Amount Contacted (l/day)	Amount Ingested (l/day)	Dermal Absorption Fraction	Ingestion Absorption Fraction	Frequency of Contact (days)	Fraction of Year Exposed	Years of Exposure	Fraction of Lifetime Exposed	Body Weight (kg)	Dermal Intake (l/kg/day)	Ingestion Intake (l/kg/day)	Target Level (ug/l)
TCE	3.33E-01	3.33E-07	1.10E-02	3.03E-05	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.56E-02	11.0E+00
1,1-DCE	3.33E-01	3.33E-07	5.80E-01	5.75E-07	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.95E-02	12.01E+02
PCE	3.33E-01	3.33E-07	5.10E-02	6.54E-06	0.00E+00	2.00E+00	0.00E+00	1.00E+00	3.65E+02	1.000	7.00E+01	1.00	7.00E+01	0.00E+00	2.86E-02	12.29E+01

INTERMEDIATE PLUME: OPTION 3

Residual Risks Associated with: TCE at MCL, 1,1-DCE and PCE at 10⁻⁶ Risk Level

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAB Potency Estimate (ug/kg/day)-1	Risk Estimate
1,1-DCE	1.00E-01	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-03	1.00E+03	5.60E-01	1.66E-06
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
PCE	1.00E-03	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	2.86E-02	1.00E+03	5.20E-02	1.49E-09
											TOTAL RISK	4.72E-06

FIGURE 2-2. INTER. PLUME CONTAMINATION

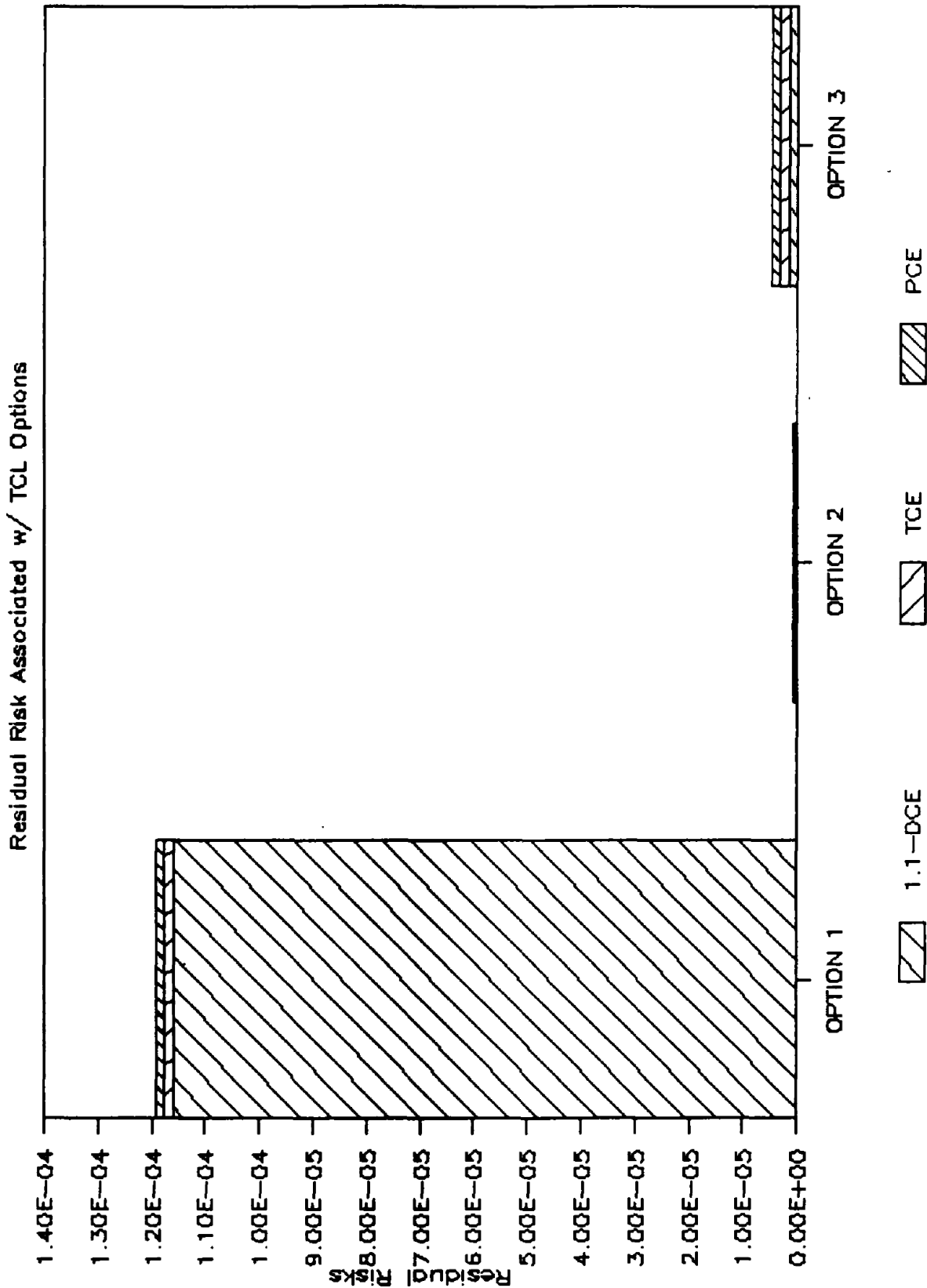


TABLE 2-6
 CADILLAC AREA GROUNDWATER CONTAMINATION
 PRELIMINARY TARGET LEVELS
 REXAIR PLUME

<u>CHEMICAL</u>	<u>(mg/l)</u>	<u>SOURCE</u>
TCE	0.005	MCL
MC	0.005	CALCULATED

The residual risks associated with preliminary target levels for TCE and MC in the Rexair plume are calculated in Table 2-7. The total residual risk is 2.64×10^{-6} . This is well within the target risk range. Therefore, the preliminary target level analyses for the Rexair plume will not be taken further.

2.3.2.4 Northernnaire Chromium Plume. The chemical of concern that poses a potentially significant risk in the Northernnaire plume is hexavalent chromium (Cr[+6]). This chemical is not carcinogenic via the ingestion route, and it has an applicable health-based ARAR, a final MCL of 0.05 mg/l. Therefore, the preliminary target level for Cr(+6) is set at its MCL.

2.3.3 Final Groundwater Target Levels

(TO BE ADDED AFTER MEETING WITH MDNR AND EPA)

2.4 TARGET LEVELS FOR SOIL REMEDIATION

The general response objective for remediation of soils at the Cadillac Industrial Park is to reduce residual contamination to levels that will no longer significantly contribute to the groundwater contamination. This section describes the approach to development of soil target levels, and presents the target levels for soils in both the Kysor and the Four Winns areas.

TABLE 2-7

Residual Risk Associated with Target Cleanup Levels
 Cadillac Area Groundwater Contamination
 Ground Water Ingestion

REPAIR PLUME

Residual Risks Associated with Groundwater Contaminants at their Preliminary Target Levels

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAB Potency Estimate (ug/kg/day)-1	Risk Estimate
MC	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	7.50E-03	1.07E-06
TCE	5.00E+00	2.00E+00	1.00E+00	7.00E+01	3.65E+02	1.00	7.00E+01	1.00	1.43E-01	1.00E+03	1.10E-02	1.57E-06
											TOTAL RISK	2.64E-06

2.4.1 Approach

The first step in the development of soil target levels is to identify the chemicals in each area that are or have the potential to significantly impact groundwater. This was accomplished for Kysor soils by identifying the chemicals of concern that (1) are present in the soils at high concentration, and (2) are posing "significant risks (or are at levels above ARARs) in the shallow plume. Using these criteria, TCE, toluene, and 1,1-TCA are included in this TCL evaluation. Xylene was also included because it was found at the highest concentrations in the Kysor soils. Inadequate current data are available to assess Four Winns soils. However, soil TCLs are developed for those chemicals which may be in the soils at significant concentrations based on 1985 and very limited 1987 data. These include 1,1,1-TCA, toluene, TCE, and MC in the Four Winns soils.

There are currently no health-based ARARs for these chemicals in soils.

Therefore, all preliminary soil target levels were calculated on the basis of risk. This was accomplished by using the preliminary groundwater target levels identified for these chemicals in Section 2.3 and back calculating to a "safe" concentration in the soil using the defining equation for the USEPA Organic Leaching Model (OLM). This equation is as follows:

$$C_w = 0.00221 \times C_s^{0.678} \times S^{0.373}$$

Where:

C_w = Preliminary target level in water

C_s = Preliminary target level in soil

S = Contaminant's water solubility at ambient temperature

These preliminary soil target levels are revised in the Final Target Level section to reflect any revisions that are made to the preliminary groundwater target levels.

2.4.2 Development of Preliminary Soil Target Levels

The following subsections summarize the identification of preliminary target levels for soils in the Cadillac Industrial Park.

2.4.2.1 Kysor. The parameters necessary to calculate preliminary target levels for chemicals in the Kysor soils, and the TCLs themselves, are summarized in Table 2-8. Xylene did not have a preliminary groundwater target level, as it was not identified as posing a significant risk in the baseline RA. Therefore, a health-based groundwater TCL is calculated for xylene in Appendix B. Preliminary TCLs for Kysor soils range from 0.07 ppm for TCE to 244,489 ppm for xylene.

2.4.2.2 Four Winns. Table 2-9 summarizes the parameters and preliminary TCLs for chemicals in the Four Winns soils. Preliminary soil TCLs range from 0.02 ppm for MC to 724 ppm for toluene.

2.4.3 Final Soil Target Levels

(TO BE ADDED AFTER MEETING WITH MDNR AND EPA)

TABLE 2-8

CADILLAC AREA GROUNDWATER CONTAMINATION
PRELIMINARY TARGET LEVELS
KYSOR SOILS

<u>CHEMICAL</u>	<u>GW TARGET LEVEL</u> (mg/l)	<u>SOLUBILITY</u> (S, mg/l)	<u>SOIL TARGET LEVEL</u> (MG/KG)
TCE	0.005	1100	0.07
TOLUENE	2	535	724
XYLENE	69	180	244489
1,1,1-TCA	0.2	4400	7.6

Use
Proposed

mg/l in water to determine levels in soil

141 g/m target level

TABLE 2-9
 CADILLAC AREA GROUNDWATER CONTAMINATION
 PRELIMINARY TARGET LEVELS
 FOUR WINNS SOILS

<u>CHEMICAL</u>	<u>GW TARGET LEVEL (mg/l)</u>	<u>SOLUBILITY (S, mg/l)</u>	<u>SOIL TARGET LEVEL (MG/KG)</u>
TCE	0.005	1100	0.07
1,1,1-TCA	0.2	4400	7.6
TOLUENE	2	535	724
MC	0.005	16700	0.02

2.5 TREATED GROUNDWATER DISCHARGER REQUIREMENTS

Potential options for the disposal of treated groundwater include discharge to groundwater, surface water, or a POTW. Discharge requirements for these options are identified in Table 2-10, using the chemicals identified in the RA as contributing the most to the total risk in each plume. Although other contaminants were detected in groundwater in past sampling episodes, remediation of the specified compounds identified from the RA to discharge limits should remove other compounds that may be present to acceptable levels.

While the SDWA's MCLs and MCLGs and the Clean Water Act's (CWA's) AWQC establish clean-up levels for public drinking water systems, Michigan's Water Resources Commission has established groundwater quality rules for discharges to groundwater, under Michigan Groundwater Quality Rules (MGWQ), Part 22. Generally, these discharge rules are more stringent than MCLs and AWQC, and maintain that the quality of groundwater in all usable aquifers shall not be degraded from local background groundwater quality as the result of a discharge. Compounds at concentrations that exceed MCLs shall not be discharged into groundwater in usable aquifers even in those cases where the local background groundwater levels for these compounds exceed the MCL levels. Generally, Michigan's nondegradation policy for groundwater discharges prohibits discharges of organic contaminants at levels above detection. Variances may be granted by the commission for exceptional circumstances where it is determined that: strict conformity with the rule is not economically or technically feasible, a prudent alternative does not exist, and such variance is consistent

TABLE 2-10

TREATED GROUNDWATER DISCHARGE REQUIREMENTS

Compound	Maximum Concentration Detected (mg/l)	Discharge to GW (mg/l)	Reference	Discharge to Surface Water (mg/l)	Reference	Discharge to Cadillac POTW ¹ (mg/l)
1,1,1-trichloroethane	32.0	0.2*	MCL	18.4	AWQC***	.1
trans-1,2-dichloroethylene	4.6	0.07*	Proposed MCLG	.00185	AWQC****	.1
1,1-dichloroethylene	1.0	0.007*	MCL	.00185	AWQC****	.1
1,2-dichloroethane	2.0	0.005*	MCL	.243	AWQC****	.1
methylene chloride	0.69	0.005	Calculated			.1
tetrachloroethylene	0.72	0.001	Calculated	.00885	AWQC****	.1
trichloroethylene	153.0	0.005*	MCL	.0807	AWQC****	.1
chromium (hexavalent)	0.773	0.05*	MCL	.011	AWQC**	.355 (total chromium)

* A nondetectable level may be required under MGWQ, Part 22

** AWQC - Chronic

*** AWQC - Water and Fish Ingestion

**** AWQC - Fish Consumption Only

¹ Reference - Cadillac POTW Pretreatment Program Requirements

with the promotion of the public health, safety, and welfare. Therefore, requirements for discharges to groundwater, presented in Table 2-10, are based on MCLs, target levels developed for a 10^{-6} risk, or nondetectable levels.

Michigan Water Quality Standards and National Pollutant Discharge Elimination System (NPDES) requirements for surface water discharges are decided on a case-by-case basis by the appropriate MDNR division. MDNR is presently developing limits for the compounds listed in Table 2-10 for a potential discharge to the Clam River. In the absence of specific standards, AWQC are used as guidance in establishing levels, so that the appropriate extent of groundwater treatment needed before discharging to surface water may be determined.

The selected water discharge limits, presented in Table 2-10, are extremely conservative because the effects of dilution have not been considered. For example, AWQC values are criteria that should not be exceeded for the receiving body of water, but may be exceeded by the waste stream. The levels for surface water discharge will be updated as more information is obtained from MDNR.

Michigan Water Quality Standards also require a minimum of 5 milligrams per liter of dissolved oxygen to be maintained in inland waters, unless the water body is a specific inland lake or protected for coldwater fish, for which more stringent levels will be set.

If groundwater is discharged to a POTW, it must meet pretreatment standards set by the POTW. The City of Cadillac's POTW does have an approved pretreatment program with specific limits for the contaminants listed in Table 2-10. If the

Cadillac POTW is utilized for disposal, the groundwater will be pretreated to meet appropriate standards.

GLOSSARY OF ACRONYMS

ARARs	Applicable or Relevant and Appropriate Requirements
AWQC	Ambient Water Quality Criteria
CAG	Carcinogen Assessment Group
CDI	Chronic Daily Intake
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (the Superfund statute)
CWA	Clean Water Act
1,2-DCA	1,2-dichloroethane
1,1-DCE	1,1-dichloroethylene
FS	Feasibility Study
MCL	Maximum Contaminant Level
MCLG	Maximum Contaminant Level Goal
MDNR	Michigan Department of Natural Resources
MGWQ	Michigan Groundwater Quality Rules
NCP	National Contingency Plan
NPDES	National Pollutant Discharge Elimination System
OLM	Organic Leaching Model
PCE	tetrachloroethylene
POTWs	Publicly Owned Treatment Works
ppm	parts per million
RA	Risk Assessment
RI	Remedial Investigation
ROD	Record of Decision
SARA	Superfund Amendments and Reauthorization Act of 1986
SDWA	Safe Drinking Water Act

TBCs	To Be Considered Materials
1,1,1-TCA	1,1,1-trichloroethane
TCE	trichloroethylene
TCL	target clean-up levels
USEPA	United States Environmental Protection Agency
VOCs	Volatile Organic Compounds

APPENDIX A
CHEMICAL-SPECIFIC ARARS

APPENDIX A
CHEMICAL-SPECIFIC ARARS
CADILLAC AREA FS

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
<u>GROUNDWATER</u>			
Federal Applicable Requirements	SDWA - Maximum Contaminant Levels (MCLs) (40 CFR 141.11 - 141.16)	MCLs have been promulgated for a number of common organic and inorganic contaminants. These levels regulate the concentration of contaminants in public drinking water supplies, and are considered applicable for groundwater aquifers used for drinking water.	When the risks to human health due to consumption of groundwater are assessed, concentrations of contaminants should be compared to their MCLs.
Federal Relevant and Appropriate Requirements	SDWA - Maximum Contaminant Level Goals (MCLGs)	MCLGs are health-based criteria that are relevant and appropriate for drinking water sources, as a result of SARA, when extraordinary risks exist. These goals are available for a number of organic and inorganic contaminants.	Groundwater concentrations of noncarcinogenic contaminants should be compared to their MCLGs when extraordinary risks exist.
	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC should be used in characterizing human health risks due to contaminant concentrations in groundwater.
State Applicable Requirements	HWRL, Act 245, R 323 Part 22, Groundwater Discharge Rules	Draft proposed rules to establish allowable discharge levels to protect Michigan groundwater quality in all usable aquifers as potential water supplies.	The draft revisions to the existing Part 22 Rules are still in the approval process; it is unknown when they will take effect.
Federal Advisories and Guidance to be Considered	Health Advisories (USEPA Office of Drinking Water)	Health advisories are estimates of risks due to consumption of contaminated drinking water; they consider noncarcinogenic effects only.	Health advisories should be considered for contaminants in groundwater that is used for drinking water.
	USEPA Risk Reference Doses (RfDs)	RfDs are dose levels developed by USEPA for noncarcinogenic effects.	USEPA RfDs are used to characterize risks due to exposure to contaminants in groundwater, as well as other media.
	USEPA Carcinogen Assessment Group Potency Factors	Potency Factors are developed by USEPA from Health Effects Assessments (HEA) or evaluations by the Carcinogen Assessment Group.	USEPA Carcinogenic Potency Factors are used to complete the individual incremental cancer risk resulting from exposure to contaminants.

APPENDIX A (continued)
CHEMICAL-SPECIFIC ARARS
CADILLAC AREA FS

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE RI
	Acceptable intake - Chronic (AIC) and Subchronic (AIS) - USEPA Health Assessment Documents	AIC and AIS values are developed from RfDs and HEAs for noncarcinogenic compounds.	AIC and AIS values are used to characterize the risks due to several noncarcinogens in various media.
	USEPA Office of Water Guidance - Water Related Fate of 129 Priority Pollutants (1979)	This guidance manual gives transport and fate information for 129 priority pollutants.	The manual is used to assess the transport and fate of a variety of contaminants.
<u>DISCHARGE TO PUBLICLY OWNED TREATMENT WORKS</u>			
Federal Applicable Requirements	CWA - Pretreatment Standards (40 CFR 403)	Discharges to a POTW must comply with the POTW's USEPA-approved pretreatment requirements.	POTWs in the area with approved pretreatment programs will be identified. The discharge must be treated to those levels required by the program.
State Applicable Requirements	MWRL, Act 245, R 323.2163 Pretreatment Program	Discharges to a POTW must comply with the POTW's pretreatment requirements.	POTWs in the area with approved pretreatment program will be identified. The discharge must be treated to those levels required by the program.
<u>DISCHARGE TO SURFACE WATER/CLAM RIVER</u>			
State Applicable Requirements	Michigan Water Quality Standards, Act 245, R 323	Standards have been developed for dissolved oxygen; temperature standards for specific contaminants are developed on a case-by-case basis.	These standards will be used for setting limits for discharge to surface water.
Federal Relevant and Appropriate Requirements	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health-based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC may be used for setting limits for discharge to surface water.

APPENDIX B
CALCULATIONS FOR
PRELIMINARY TARGET LEVELS
CADILLAC AREA GROUNDWATER CONTAMINATION

DISK: CADFS
FILE: CADCARL

TABLE B-1
Preliminary Groundwater Target Levels for Carcinogens
Lifetime Consumption
CADILLAC GROUNDWATER CONTAMINATION
CADILLAC, MI

The equations used to calculate target levels are:

$$\text{Target Risk For Individual Compound} = 10^{-6} \times \text{Risk Apportionment Factor (AF)}$$

$$\text{Target Chronic Daily Intake (CDI: } \mu\text{g/kg/day)} = \frac{\text{Target Risk}}{\text{CAB Potency Factor (} \mu\text{g/kg/day)}^{-1}}$$

$$\text{Target Level (} \mu\text{g/L)} = \frac{\text{Target CDI (} \mu\text{g/kg/day)}}{\text{Human Intake Factor (L/kg/day)}}$$

$$\text{where: Human Intake Factor} = \frac{\text{Amount of water Consumed Per Day}}{\text{Average Adult Body Weight}} = \frac{2 \text{ L/day}}{70 \text{ kg}} = 0.029$$

Compound	AF	Target Risk	Potency Factor ($\mu\text{g/kg/day)}^{-1}$	Target CDI ($\mu\text{g/kg/day)}$	Human Intake Factor (L/kg day)	Target Level ($\mu\text{g/L)}$
SHALLOW & INTER. PLUMES:						
TETRACHLOROETHENE	1.00E+00	1.00E-06	5.18E-02	1.96E-05	0.029	5.76E-04
DEKAIRE PLUME:						
METHYLENE CHLORIDE	1.00E+00	1.00E-06	7.50E-03	1.33E-04	0.029	4.60E-03

TABLE B-2
Preliminary Groundwater Target Levels for Noncarcinogens
Lifetime Consumption
CADILLAC GROUNDWATER CONTAMINATION
CADILLAC, MI

$$\text{Target Chronic Daily Intake (CDI): mg/kg/day} = \text{Target Risk Level} \times \text{RfD}$$

$$\text{Target Level (ng/L)} = \frac{\text{Target CDI (ng/kg/day)}}{\text{Human Intake Factor (L/kg/day)}}$$

$$\text{where: Human Intake Factor} = \frac{\text{Amount of Water Consumed Per Day}}{\text{Average Adult Body Weight}} = \frac{2 \text{ L/day}}{70 \text{ kg}} = 0.029 \text{ (L/kg/day)}$$

[illegible]

2

2